AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 35 · STUDENT LESSON

Capstone Train, Analyze, Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 26: a reproducible comparison identifies data, version and settings.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Run a frozen final evaluation with complete accounting. Try the recall before opening notes.

Your learning target

I can run a frozen **final evaluation** with complete accounting.

I can separate error analysis from an untouched-test claim.

I can document revisions and their **evidence status**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

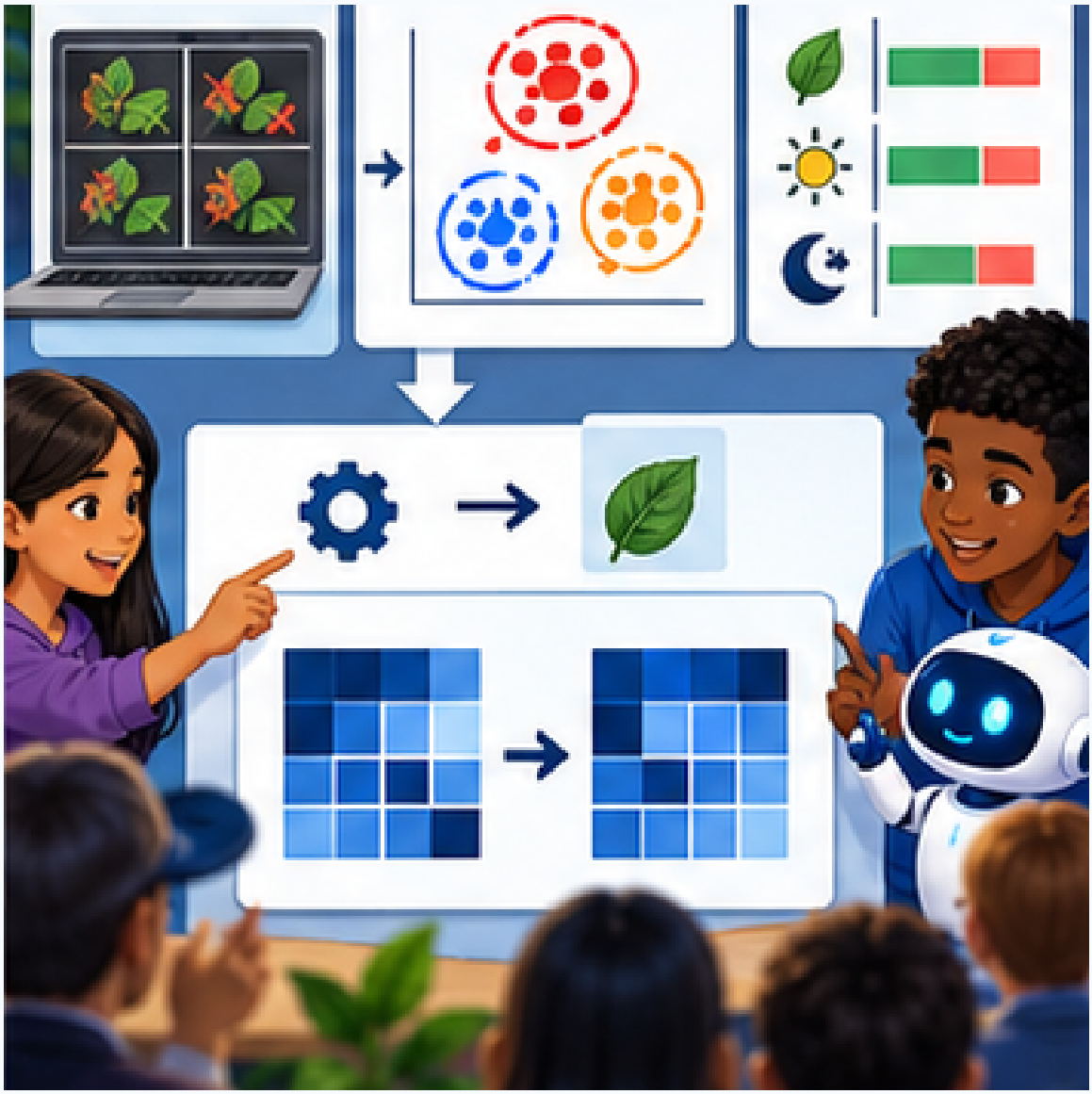
Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.



Your target

LEARNING TARGET A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

Builder name and experiment date:

Readiness for the connected explanation

What changes after final-test labels inform a revision?

This week's end goal

By the end, I can explain capstone train, analyze, improve using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

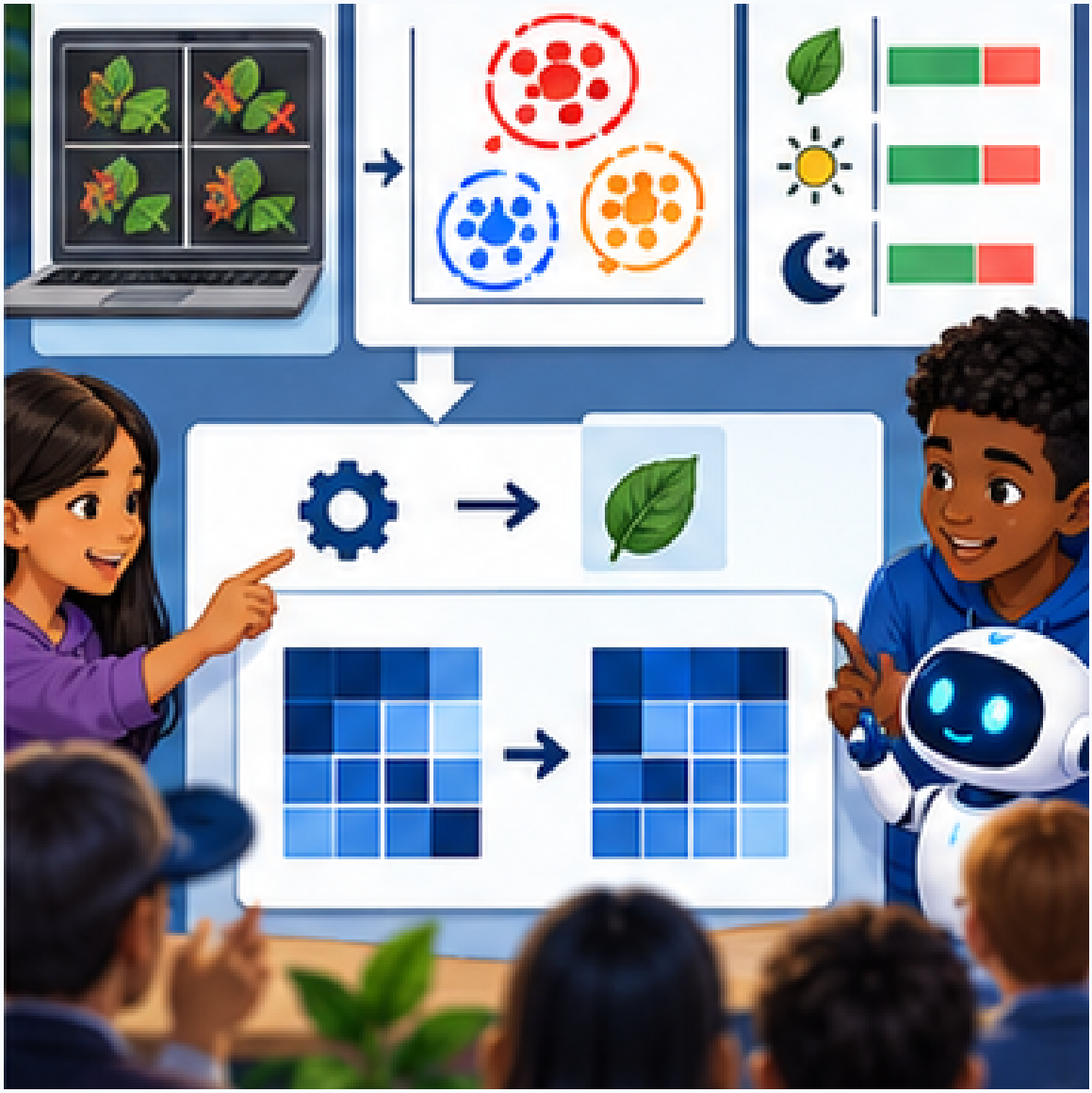
Notice the evidence

> Investigate the supplied Capstone Train, Analyze, Improve case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use capstone train, analyze, improve to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

The teacher releases final-test labels only after checking the freeze record. Score baseline and candidate on the same planned rows. Save every **prediction**, denominator and failure.

Do not edit a model while reading labels and then present only the edited version as the original test.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone evaluation and revision.

STUDENT LESSON · 3 OF 18

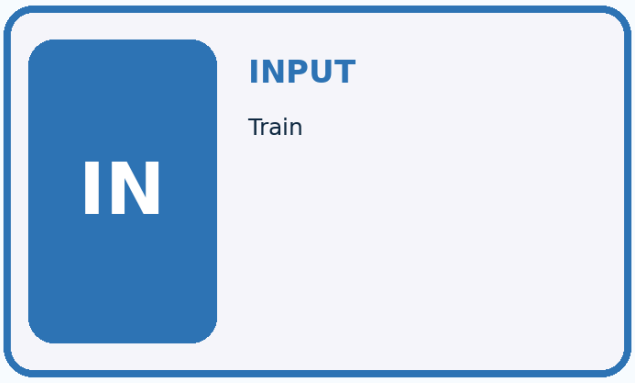
See how it works

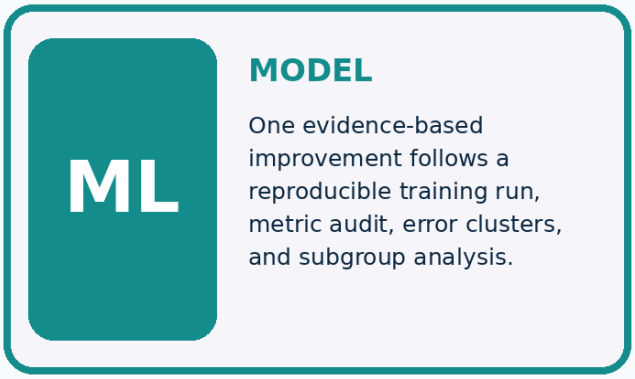
> Follow the mechanism for Capstone Train, Analyze, Improve. Connect each labelled input, deciding step and result.

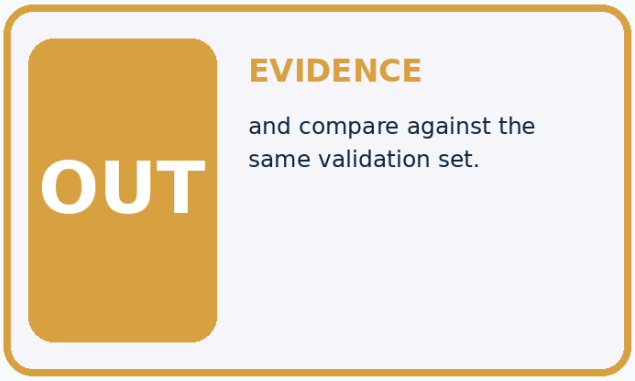
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Train, audit, change one factor, and compare against the same validation set.

MODEL IDEA One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Model v1 passes 14/20 validation cases; evening accuracy is 40%. The six failed cases are low-light cases.

One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Illustrative frozen test has 12 rows: candidate[[7, 1],[2, 2]], baseline[[8, 0],[4, 0]], blocked positive. Candidate accuracy 9/12=75 percent, precision 2/3 and recall 2/4; baseline accuracy 8/12 about 66.7 percent.

The candidate adds one false alarm and misses two rather than four blocked cases.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

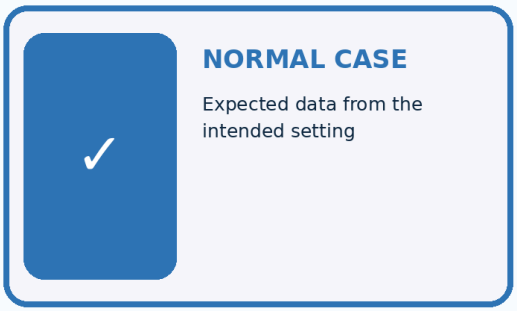
Compare the cases

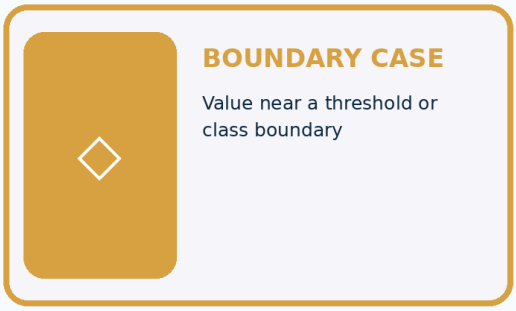
> Compare the supplied cases for Capstone Train, Analyze, Improve. Identify the detail that changes the conclusion.

EVALUATION DESIGN

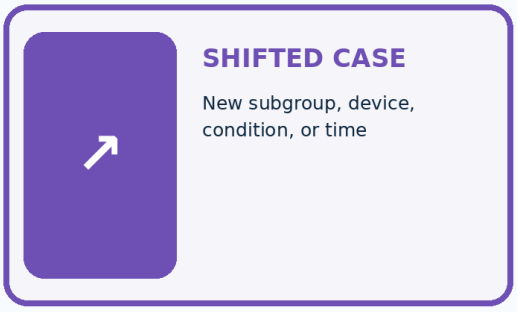
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: A revision fixes one miss but creates two extra false alarms.

Predicted result

Prediction: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

Why the result follows

Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Capstone Train, Analyze, Improve to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
BASELINE RUN
first reproducible model result
2
ERROR AUDIT
systematic review of mistakes
3
BEFORE/AFTER
same evaluation around one change
4
REGRESSION
previously passing case now fails

Term	Precise meaning	My ML example
baseline run	first reproducible model result	
error audit	systematic review of mistakes	
before/after	same evaluation around one change	
regression	previously passing case now fails	

MISCONCEPTION TO REPAIR A higher overall score proves every important aspect improved.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Final evaluation	Assessment of the frozen procedure on held-out evidence
Error audit	A documented inspection of mistakes and missing predictions
Revision log	A record of changes, reasons and affected evidence
Evidence status	Whether a result is development, untouched test or post-test reuse

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone Train, Analyze, Improve, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 35: Capstone Train, Analyze, Improve

STEP	EVIDENCE AND REASONING
1	Model v1 passes 14/20 validation cases; evening accuracy is 40%.
2	The six failed cases are low-light cases. One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.
3	Add reviewed low-light training examples while freezing features and model settings.
4	Model v2 passes 17/20; evening rises to 70%; inspect every old pass and record remaining limits.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Step 1

STEP 1 Model v1 passes 14/20 validation cases; evening accuracy is 40%.

Step 2

STEP 2 The six failed cases are low-light cases. One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

Step 3

STEP 3 Add reviewed low-light **training** examples while freezing features and model settings.

Step 4

STEP 4 Model v2 passes 17/20; evening rises to **70%**; inspect every old pass and record remaining limits.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Capstone evaluation and revision at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Frozen version	Saved before labels	Candidate v 1
First test	9 correct of 12	Report full error table
Proposed revision	Uses observed errors	Needs fresh evaluation

Compare the result with validation without claiming they should be identical. Check error types and coverage. A one-case accuracy difference on twelve rows supports a cautious classroom comparison, not a general reliability guarantee. Keep the original prediction ledger unchanged.

What does the one-correct-case improvement establish? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Train, Analyze, Improve

Choose the next capstone revision from explicit false alarms and misses.

The facts we will use

Supplied facts

Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/35

Candidate, before revision

	predicted spotted	predicted healthy
actual spotted	4	2
actual healthy	1	9

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/35

Capstone Train, Analyze, Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare accuracy

Candidate13/16=81.25%; baseline10/16=62.5%.



2. Compare detection

Candidate catches4/6 spotted leaves; baseline catches0/6.



3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

CHANGE ONE THING

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare accuracy	Candidate13/16=81.25%; baseline10/16=62.5%.
2. Compare detection	Candidate catches4/6 spotted leaves; baseline catches0/6.
3. Inspect errors	Review candidate’s two misses and one false alarm with conditions and label evidence before revising.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Keep the stated limit

Read the labelled picture: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A revision fixes one miss but creates two extra false alarms. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone Train, Analyze, Improve claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A higher overall score proves every important aspect improved. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A higher overall score proves every important aspect improved.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Submit the frozen test report, **error audit**, revision proposal and updated model card. A revision proposal may conclude that more appropriate data or a simpler baseline is preferable.

If no fresh test is available, stop at a transparent proposal instead of claiming the revision has been independently validated.

What must remain visible when version 2 is proposed without a fresh test?

STUDENT LESSON · 8 OF 18

Practise together

> Practise Capstone Train, Analyze, Improve with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If dim examples account for both misses, record that **observation** and inspect collection conditions. It is a hypothesis for investigation, not proof that lighting caused the failures.

A proposed feature or model change creates version 2. Scores on the same inspected test must be labelled post-test development.

Practise with less help: Audit a difference

Use workbook W2 for the connected example “Capstone evaluation and revision”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

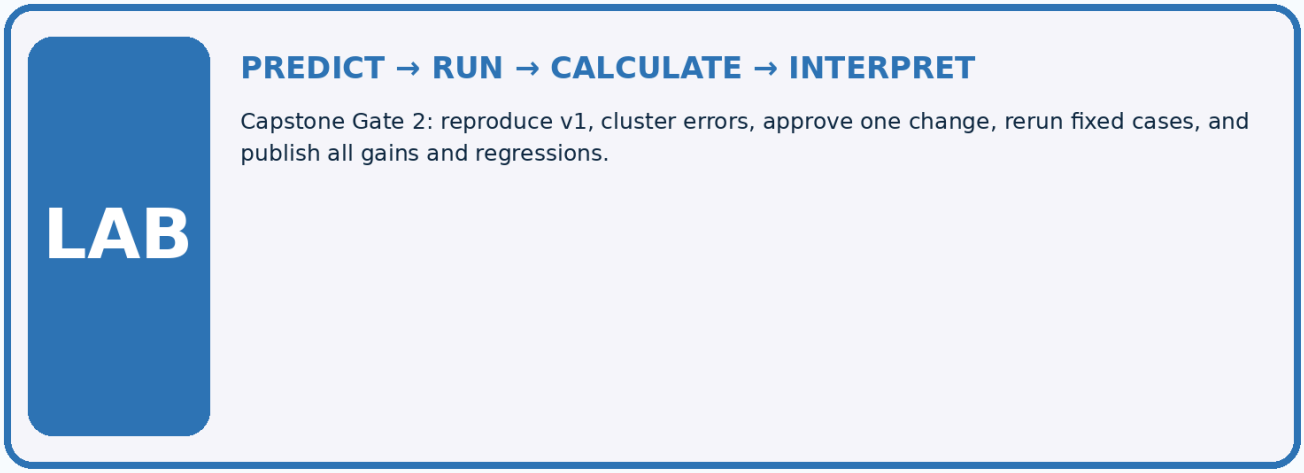
Run the investigation

> Investigate Capstone Train, Analyze, Improve. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Capstone Gate 2: reproduce v1, cluster errors, approve one change, rerun fixed cases, and publish all gains and **regression**s.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

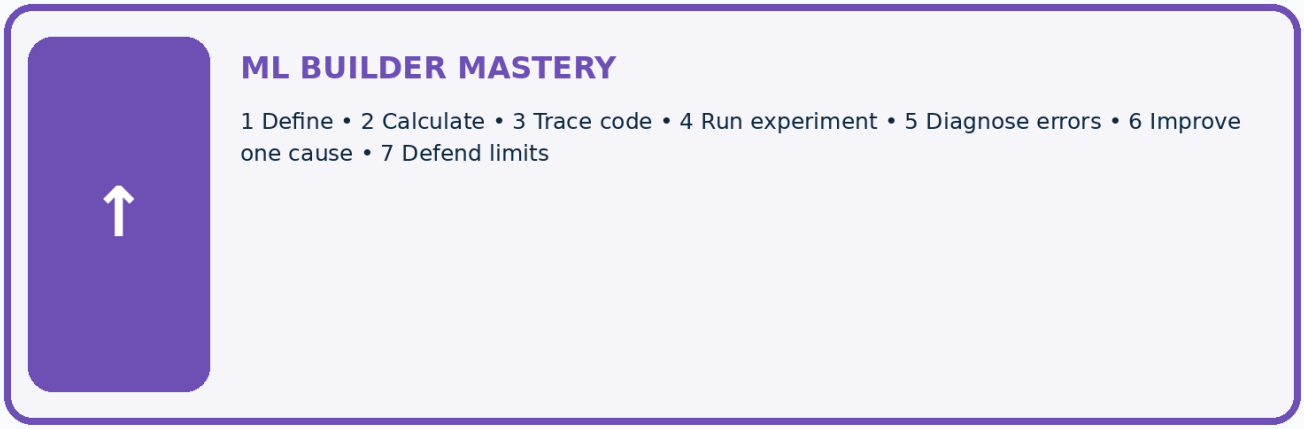
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone Train, Analyze, Improve. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **baseline run**, **error audit**, **before/after**, **regression**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A higher overall score proves every important aspect improved.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

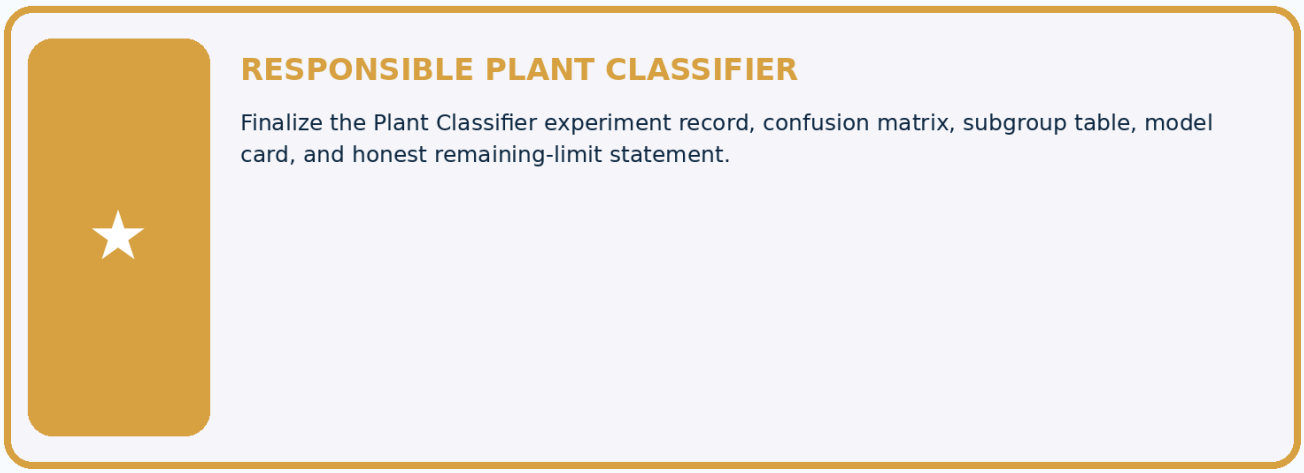
Connect your project

Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

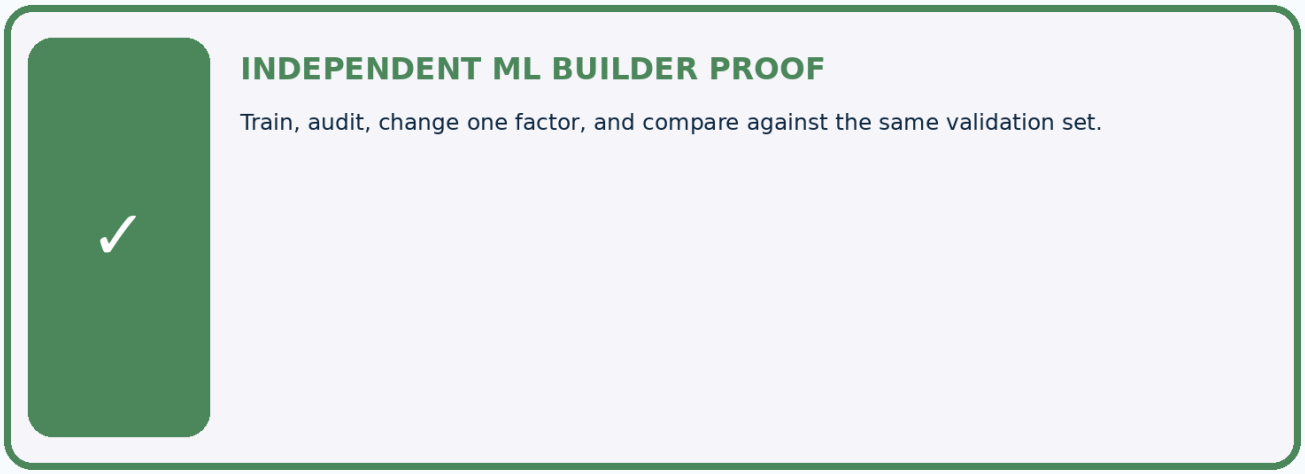
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Train, audit, change one factor, and compare against the same validation set.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: ML Builder Expo.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Capstone Train, Analyze, Improve case for supported practice. Repeated examples do not become fresh assessments.

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

The question

Plan improvement **evidence**.

Worked explanation

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep improvement evidence comparable

Evidence or stage	Value or result
Baseline	Seventy percent validation accuracy
Model	Eighty percent on the same set
Error slice	Many failures in dim images
Next revision	Change one factor and retain final test

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Plan improvement **evidence**.

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A revision raises overall validation accuracy from **80%** to 84% but lowers dim-light accuracy from **60%** to 40%. How should the improvement claim change?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone Train, Analyze, Improve, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Choose the next capstone revision from explicit false alarms and misses.

By the end, I can	How I will show it
Run a frozen final evaluation with complete accounting	Explain the deciding step, show a trace or test, and state one limit.
Separate error analysis from an untouched-test claim	Explain the deciding step, show a trace or test, and state one limit.
Document revisions and their evidence status	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone Train, Analyze, Improve without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Run a frozen **final evaluation** with complete accounting

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose the next capstone revision from explicit false alarms and misses.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

1. Inspect errors

2. Compare accuracy

3. Compare detection

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect errors Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. Read the picture in order.

Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A revision fixes one miss but creates two extra false alarms. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Repair a missing building block

Review Capstone Dataset and Baseline (ai-4/34). Goal: Define a feasible capstone question and acceptance criteria. Short practice: What provenance field prevents the crop from appearing independent?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

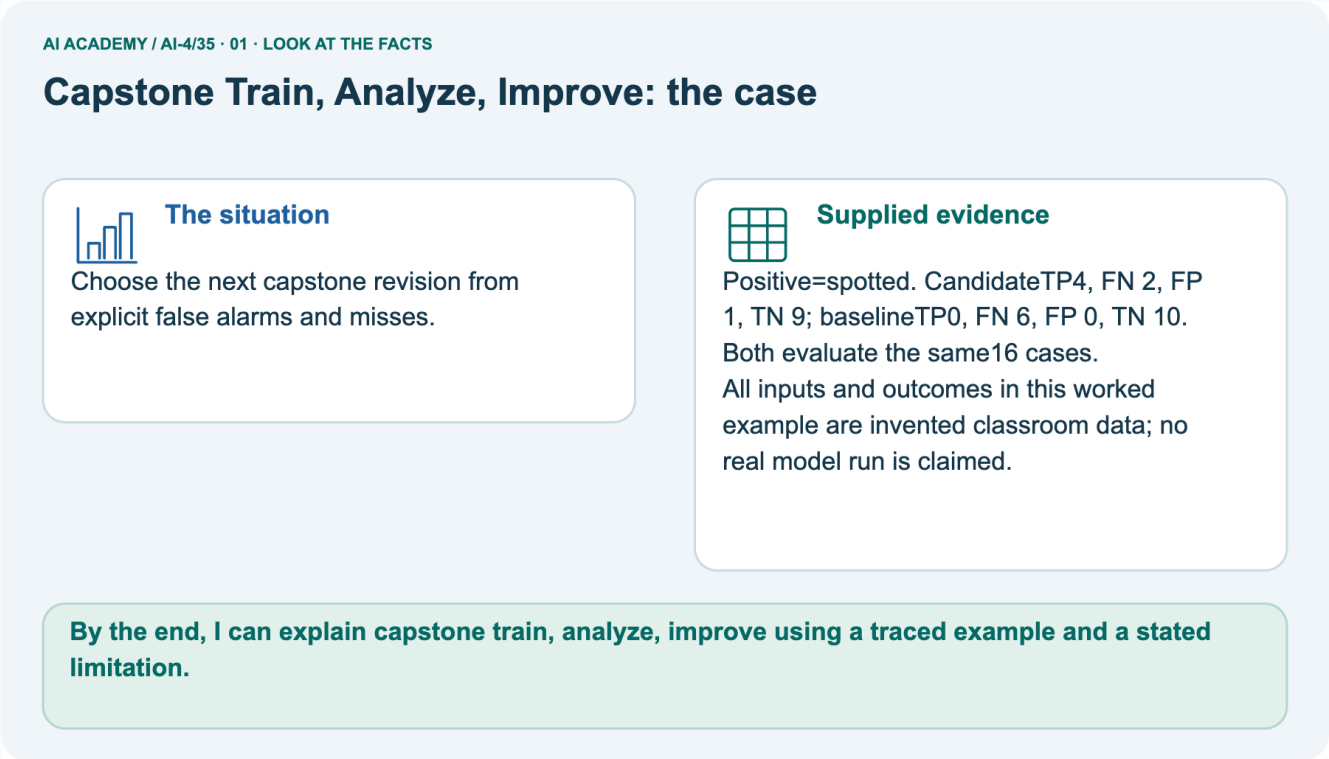
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Capstone Train, Analyze, Improve graphic using its labels, deciding evidence and relevant limitation.

01. Capstone Train, Analyze, Improve: the case



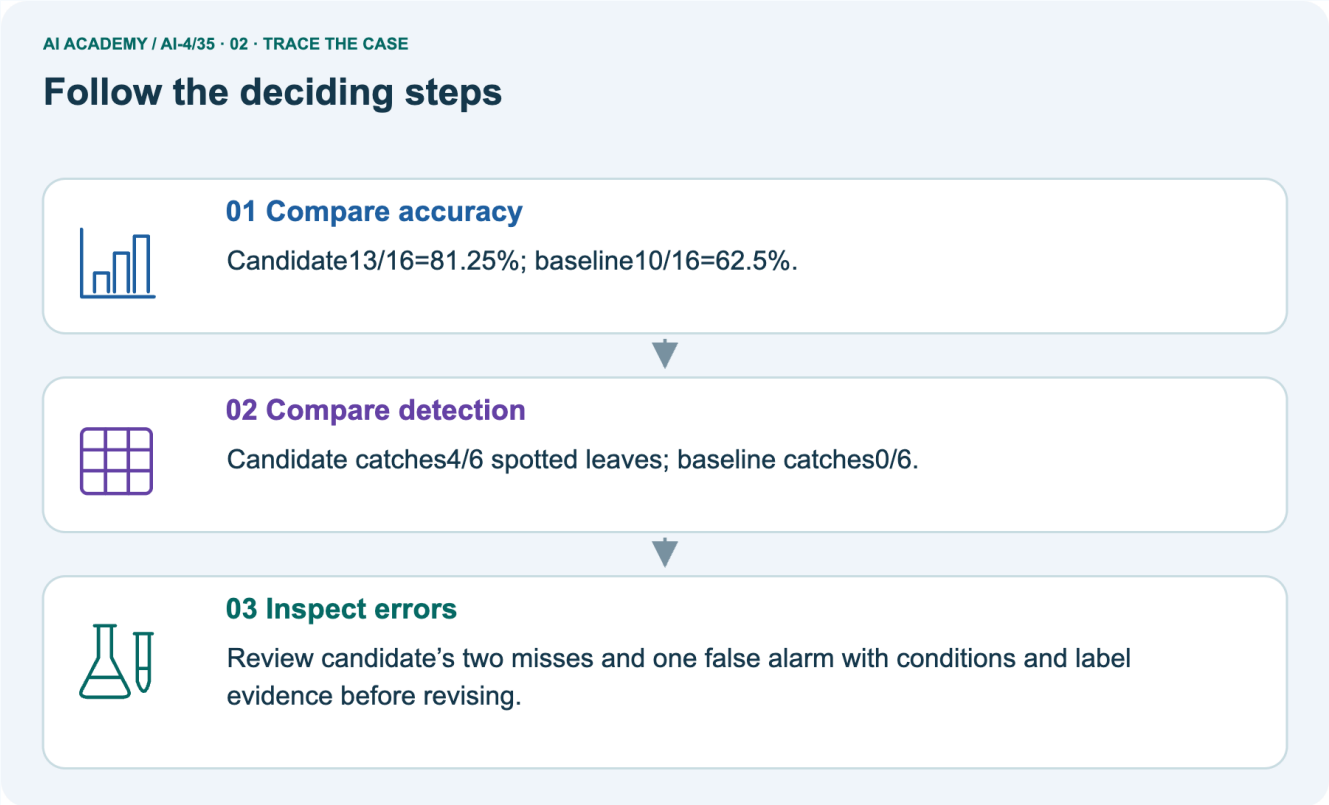
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose the next capstone revision from explicit false alarms and misses. Supplied facts: Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

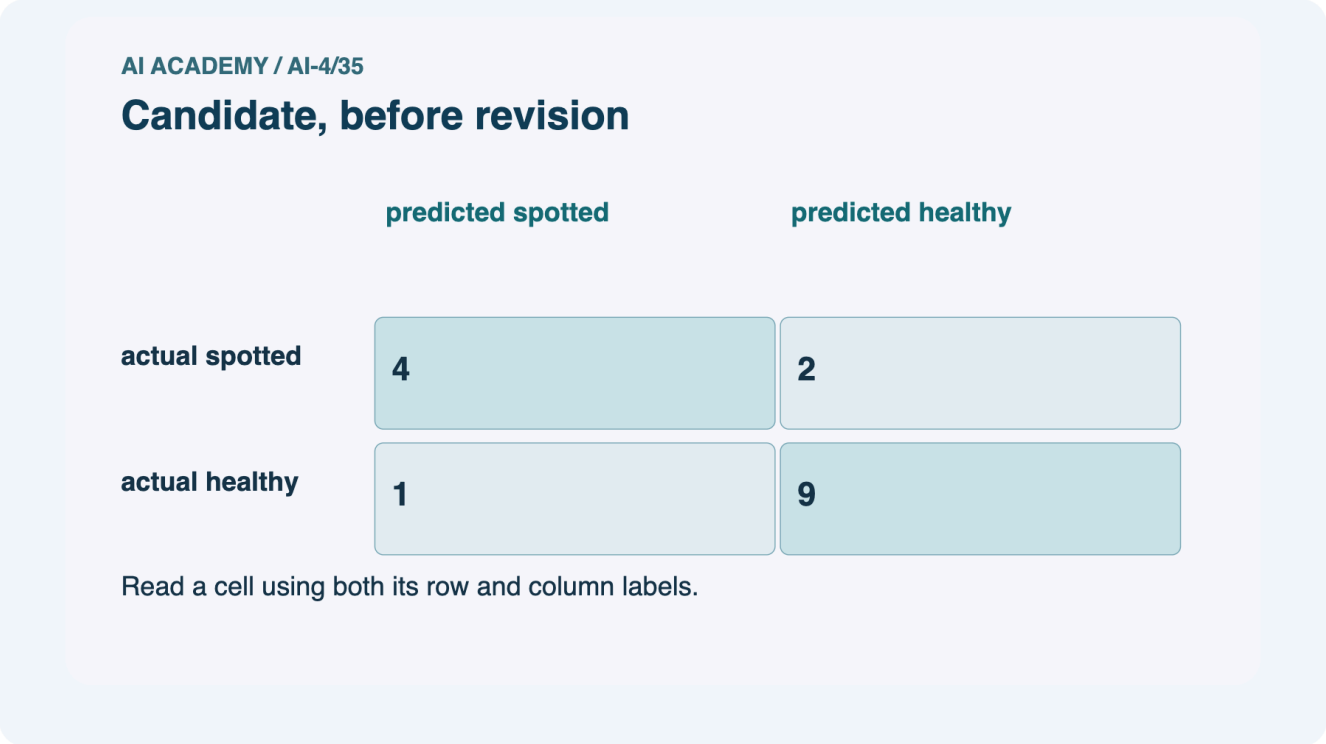


Follow the labelled picture with the matching explanation. Open the vector diagram

Compare accuracy: Candidate13/16=81.25%; baseline10/16=62.5%. → Compare detection: Candidate catches4/6 spotted leaves; baseline catches0/6. → Inspect errors: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Candidate, before revision

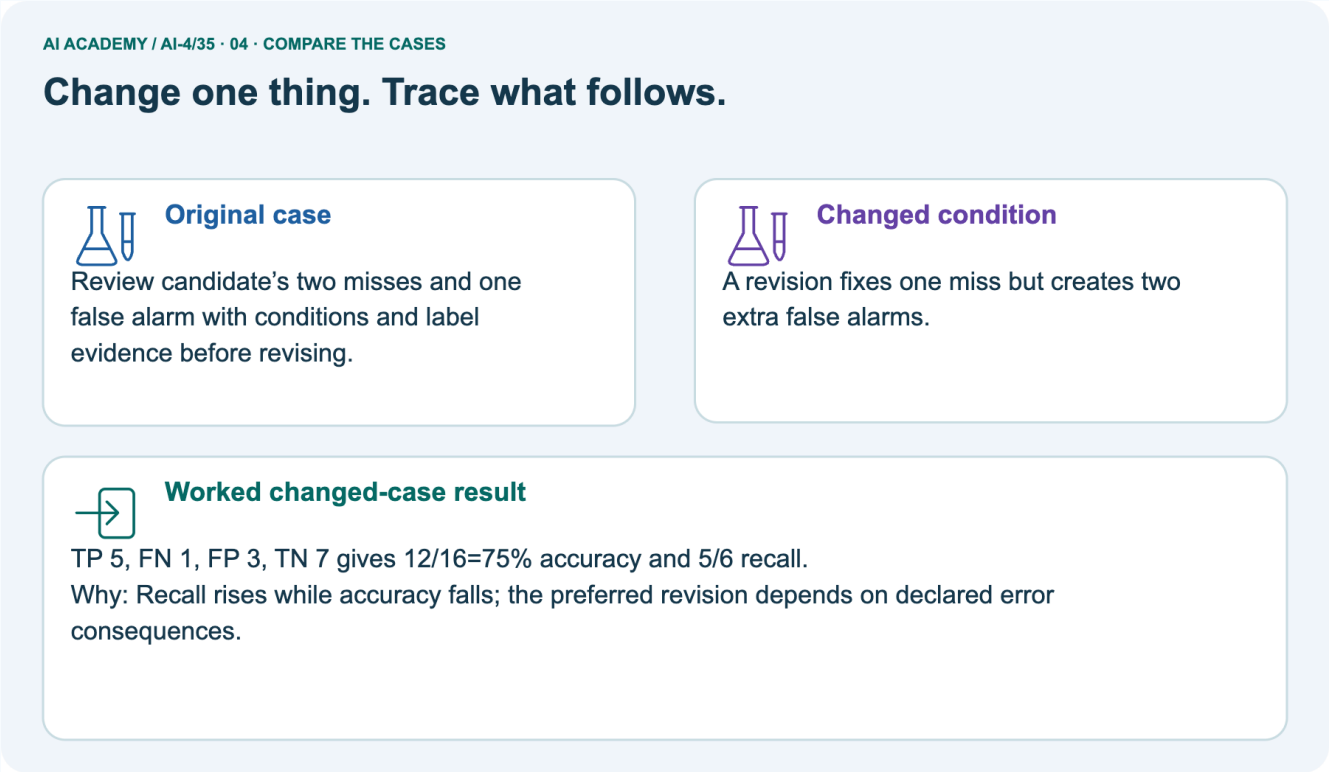


Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate, before revision. actual spotted: predicted spotted = 4, predicted healthy = 2. actual healthy: predicted spotted = 1, predicted healthy = 9.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



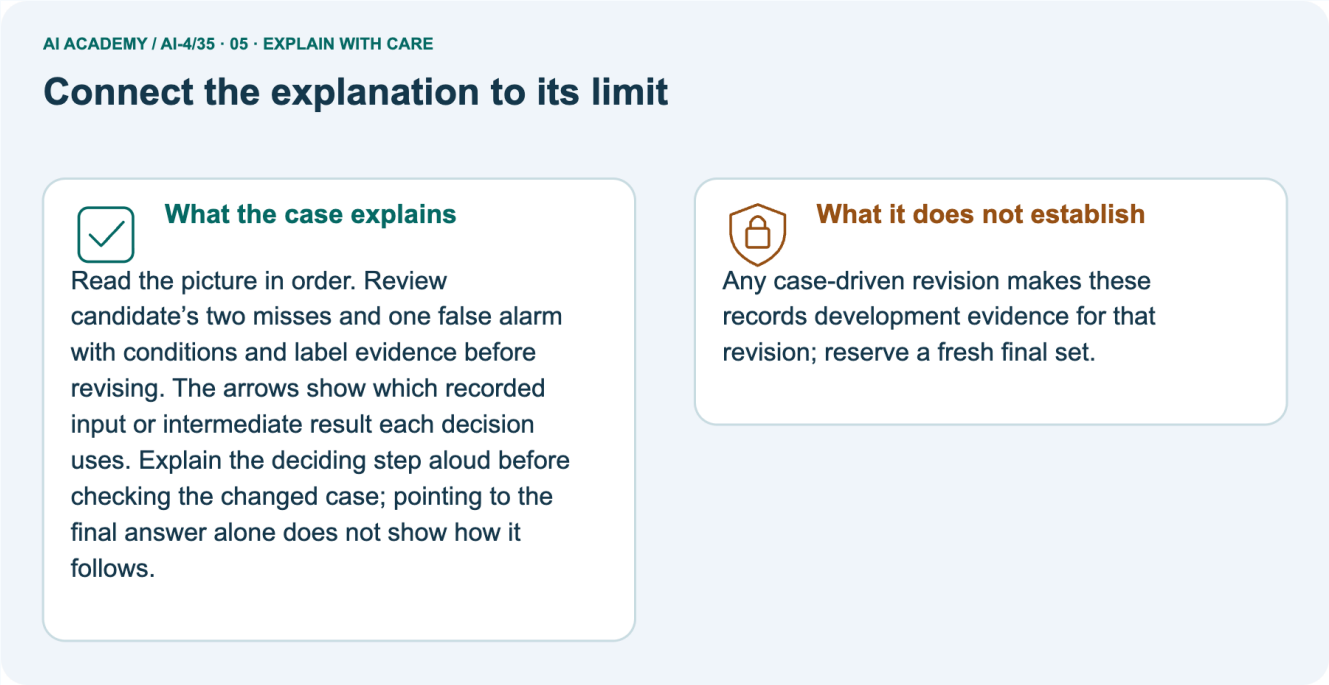
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Changed condition: A revision fixes one miss but creates two extra false alarms.

Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

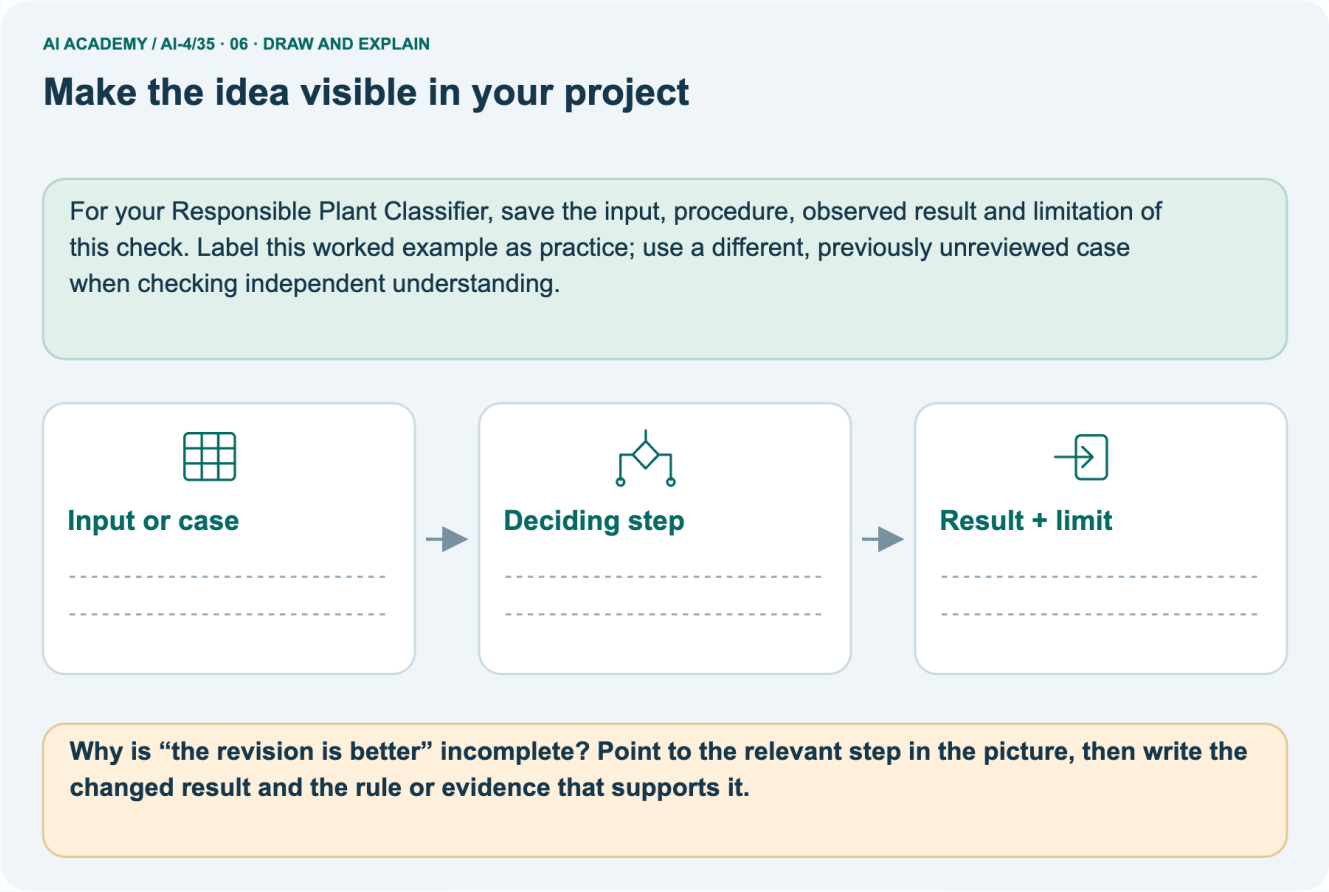
Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.